

Two Stages, One Regression

Teaching note

Beaver, Griffin, and Landsman (1982), *Journal of Accounting and Economics*

ACCTG 597A · Fall 2026 · Sam Bonsall · Penn State

■ Advertised, and cited, as a cure for collinearity

“The research design incorporates a two-stage approach which permits a determination of the incremental explanatory power of collinear variables.”

Beaver, Griffin, and Landsman (1982, p. 15)

“(The two-stage regression results) are not subject to this collinearity problem.”

Beaver and Landsman (1983, p. 66), as quoted in Christie et al. (1984, p. 205)

Lev and Ohlson (1982, p. 265, as quoted in Christie et al. 1984, p. 206) likewise wrote that the procedure appeared, on the surface, to circumvent the multicollinearity problem. Christie et al. (1984) showed within two years that the claim does not hold; the demonstration below uses BGL's own numbers.

BGL ask whether PRE adds to HC at a 0.84 correlation

SEC ASR 190 (1976–1978) required large registrants to disclose replacement cost data; BGL test whether earnings on that basis explain annual cross-sectional returns beyond historical cost earnings (313 firms; 303 in the regressions).

HC is the percentage change in historical cost EPS; PRE is the percentage change in pre-holding-gain income, in practice HC adjusted for the replacement cost depreciation difference.

BGL themselves single out the 0.84 correlation between HC and PRE as the most striking feature of Table 3 (pp. 25–26).

	HC	PRE	<i>R</i>
HC	1	0.84	0.37
PRE	0.84	1	0.30
<i>R</i>	0.37	0.30	1

Pearson correlations, 1977, from BGL Table 3 (N = 313).

BGL orthogonalize one signal and run both orderings

Panel (A)

Stage 1: $PRE_i = a + c HC_i + Z_i^A$

Stage 2: $R_i = \alpha + \beta_1 HC_i + \beta_2 Z_i^A + u_i$

Table 4, 1977: HC 0.36 ($t = 7.1$); Z^A
-0.02 ($t = -0.4$).

Panel (B)

Stage 1: $HC_i = a + d PRE_i + Z_i^B$

Stage 2: $R_i = \alpha + \beta_1 PRE_i + \beta_2 Z_i^B + u_i$

Table 4, 1977: PRE 0.16 ($t = 5.7$); Z^B
0.39 ($t = 4.2$).

"If PRE possesses explanatory power, not provided by HC, the regression coefficient (β_2) on the residual (Z) should be different from zero."

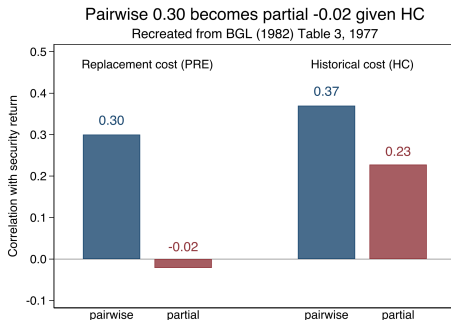
Beaver, Griffin, and Landsman (1982, p. 27)

A correlation table holds nothing fixed

Each pairwise entry in Table 3 lets the other earnings variable move. The partial correlation of PRE with R holding HC fixed follows from the three printed numbers:

$$\frac{0.30 - 0.84 \times 0.37}{\sqrt{(1 - 0.84^2)(1 - 0.37^2)}} = -0.02$$

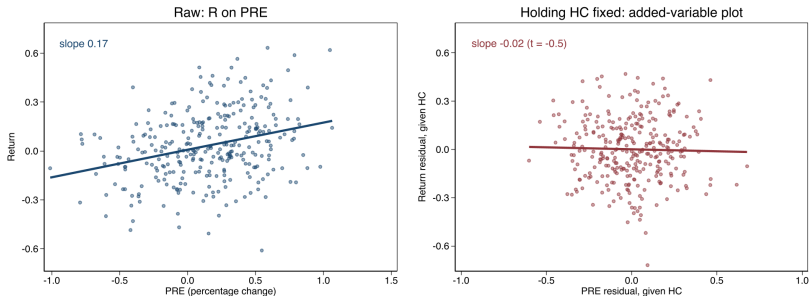
The -0.02 carries the sign of Panel (A)'s residual coefficient and, with nearly equal residual standard deviations of R and PRE, its magnitude. No entry in Table 3 satisfies *ceteris paribus*, so it cannot state what Table 4 states.



Holding HC fixed removes the entire PRE association

The PRE association vanishes once HC is held fixed

Simulated data calibrated to BGL (1982) Tables 2-4, $N = 303$



The right panel is the multiple-regression coefficient drawn as a picture (Frisch-Waugh-Lovell): both R and PRE are residualized on HC, and the slope through the residuals is -0.02 , BGL's Panel (A) result. Simulated data calibrated to Tables 2–4.

Both panels of Table 4 are one regression read twice

Start from the regression BGL never print:

$$R_i = b_0 + b_1 \text{HC}_i + b_2 \text{PRE}_i + e_i$$

Substitute the first stage, $\text{PRE}_i = a + c \text{HC}_i + Z_i^A$:

$$R_i = (b_0 + b_2 a) + (b_1 + c b_2) \text{HC}_i + b_2 Z_i^A + e_i$$

Panel (A) residual coefficient $\beta_2 = b_2$, the coefficient on PRE, with the same t -statistic. The HC coefficient $\beta_1 = b_1 + c b_2$ is the bivariate slope.

By symmetry, Panel (B) residual coefficient $\beta_2 = b_1$, the coefficient on HC; its PRE coefficient $b_2 + d b_1$ is the other bivariate slope.

Four regressions, two coefficients that matter, b_1 and b_2 ; the other two entries are bivariate slopes fixed by the first-stage slopes c and d . Christie et al. (1984, eqs. 8–9) write the same relations as $\beta_1 = \gamma_1 + \alpha_1 \gamma_2$ and $\beta_2 = \gamma_2 + \alpha_2 \gamma_1$.

Table 4 contains the one regression BGL never printed

R on HC and PRE		1977	1978	Pooled
HC	(Panel (B) residual row)	0.39 (4.2)	0.32 (4.1)	0.37 (6.3)
PRE	(Panel (A) residual row)	-0.02 (-0.4)	-0.02 (-0.4)	-0.02 (-1.1)
R^2		0.14	0.14	0.14

Recreated from BGL (1982) Table 4; t -statistics in parentheses; 303 firms. Consistency check, 1977: $0.39 + 1.46 \times (-0.02) = 0.36$, the Panel (A) HC coefficient; $-0.02 + 0.48 \times 0.39 = 0.17$, the Panel (B) PRE coefficient (printed 0.16; rounding in the two-decimal inputs).

"The two-stage procedure is equivalent to an F test on adding PRE to the regression of return (R) on historical cost earnings (HC)."

Beaver, Griffin, and Landsman (1982, p. 27, fn. 8)

A calibrated simulation reproduces the identity

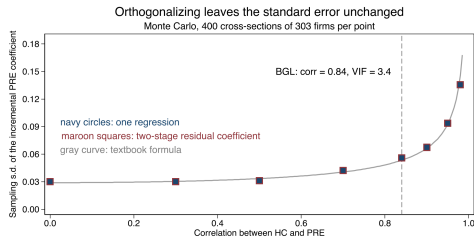
```
set seed 228
set obs 303
gen hc = 0.134 + 0.216*rnormal()
gen pre = 0.148 + 1.46*(hc-0.134) ///
          + 0.206*rnormal()
gen r = 0.018 + 0.39*(hc-0.134) ///
        - 0.02*(pre-0.148) ///
        + 0.19*rnormal()
reg pre hc           // (A) stage 1
predict zA, resid
reg r hc zA          // (A) stage 2
reg r hc pre         // one regression
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	Panel (A) <i>R</i> on HC, Z_A	Panel (B) <i>R</i> on PRE, Z_B	One regression <i>R</i> on HC, PRE
HC	0.38 (7.1)		0.42 (4.3)
PRE		0.17 (5.6)	-0.02 (-0.5)
Z_A (PRE residual, given HC)	-0.02 (-0.5)		
Z_B (HC residual, given PRE)		0.42 (4.3)	
R^2	0.14	0.14	0.14

Coefficients (t -statistics), $N = 303$; the seed was chosen so the draw approximately matches BGL's 1977 correlations, R^2 , and t -statistics. The identity checks in the log are zero to six decimals.

Orthogonalizing leaves the standard error unchanged

The two-stage residual coefficient equals the one-regression coefficient on PRE in every replication, so both share the sampling s.d. $\sigma_e / (\sqrt{N} \sigma_{\text{PRE}} \sqrt{1 - \rho^2})$. At $\rho = 0.84$ the variance inflation factor is 3.4: the standard error is 1.84 times its orthogonal value, in either procedure.



“since collinearity is inherently a data problem rather than a statistical problem, no partitions of dependent or independent variables, orthogonal or otherwise, can provide insights into the relative influence of collinear variables.”

Christie, Kennelley, King, and Schaefer (1984, p. 205)

The raw-variable coefficient is a bivariate slope

“The approach presented in panel (A) of table 4 effectively assigns all of that common explanatory power to HC.”

Beaver, Griffin, and Landsman (1982, p. 29)

“The primary cost of the two-stage procedure is the potential for errors of inference by any reader without a knowledge of the relations between the single-step and two-step procedures.”

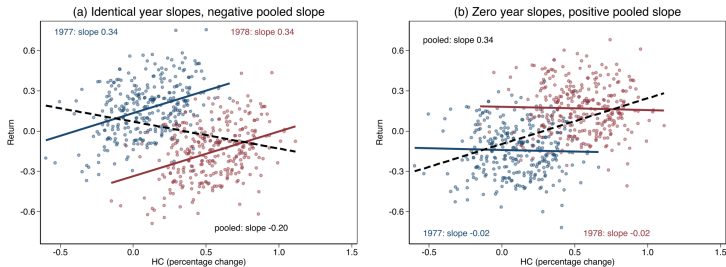
Christie, Kennelley, King, and Schaefer (1984, p. 208)

The 0.36 in Panel (A) and the 0.16 in Panel (B) are the two bivariate slopes; the assignment of the shared explanatory power is a choice of ordering, not a test.

Pooled agreement is neither necessary nor informative

Agreement between year and pooled slopes is not evidence

Simulated cross-sections, 303 firms per year



Both panels are instances of Simpson's paradox (i.e., the within-year slopes and the pooled slope differ because the year means shift). BGL's Table 4 reads 0.36, 0.29, and 0.32 for 1977, 1978, and pooled; p. 34 offers the pooled column as robustness while conceding that pooling is misspecified because residual variances differ across years.

The pooled slope is a function of the year slopes

With within-year sums of squares S_{xx}^t , within-year slopes b_t , and year means $\bar{x}_t$, $\bar{y}_t$ around the grand means $\bar{x}$, $\bar{y}$:

$$b_{\text{pooled}} = \frac{\sum_t S_{xx}^t b_t + \sum_t N_t (\bar{x}_t - \bar{x})(\bar{y}_t - \bar{y})}{\sum_t S_{xx}^t + \sum_t N_t (\bar{x}_t - \bar{x})^2}$$

With year intercepts the between-year terms vanish and $b_{\text{FE}} = \sum_t w_t b_t$ with $w_t = S_{xx}^t / \sum_s S_{xx}^s$: a weighted average of the year slopes. Without them, the pooled slope also absorbs the covariance of the year means.

What would carry information is a test of $b_{1977} = b_{1978}$ (a Chow or interaction test), which BGL do not report. The year-to-year inconsistency they themselves flag for CF and POST (p. 34) is exactly the case in which such a test has content.

Corrected within two years, reframed within eight

"In other words, BGL may be finding apparently insignificant coefficients on the replacement cost variable because their variables are collinear, rather than because the replacement cost variable is irrelevant."

Christie, Kennelley, King, and Schaefer (1984, p. 213)

"However, of the six coefficients that result from the three alternative regression specifications, only three are unique."

Jennings (1990, p. 927)

The null on PRE is a low-power test in a collinear design; the appropriate inference is a bounded effect (the effect-size-confidence logic of Burks 2023), not irrelevance.

Summary

- BGL's two-stage residual regression is numerically a single multiple regression of returns on HC and PRE. The residual coefficient in each panel is one of the two multiple-regression coefficients, and the un-orthogonalized coefficient is a bivariate slope, so the procedure adds nothing to b_1 and b_2 and provides no protection against collinearity.
- Pairwise correlations and pooled columns are not evidence about incremental content; the partial correlation implied by Table 3 is the Table 4 result, and year-by-year and pooled slopes can agree or disagree for reasons unrelated to the hypothesis.
- A design advertised as a fix can be endorsed by citation before it is checked. Christie et al. (1984) and Jennings (1990) supplied the check, and the appropriate inference from BGL's null is a bounded effect in a low-power setting rather than irrelevance.

Beaver, Griffin, and Landsman (1982) · Christie et al. (1984) · Jennings (1990) ·
github.com/sbb151/acctg597a-research-design